

ML Lab Assignment - 4

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Sem 5 – C

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Week 4: Model Selection and Comparative Analysis

Notes

1. Evaluate and compare the efficiency of own built grid search vs Scikit builtin GridSearchCV
2. Dataset: HR Attrition
3. Classification Algorithms:
 - a. Decision Tree - param_grid_dt
 - b. K Nearest Neighbours - param_grid_knn
 - c. Logistic Regression - param_grid_lr
4. Scikit Learn Pipeline : *StandardScaler* -> *SelectKBest* -> *Classifier*
 - a. StandardScaler: Standardizes features to have a mean of 0 and a standard deviation of 1.
 - b. SelectKBest: Selects the top 'k' features using a statistical test (f_classif). The number of features, k, is a key hyperparameter you will tune.
 - c. 3. Classifier: The final modeling step (Decision Tree, kNN, or Logistic Regression).

Deliverables:

1. Manual Grid Search Implementation: *run_manual_grid_search*
 - a. Define Parameter Grids
 - b. Implement Search Loop – Calculate avg ROC, AUC score for *5-fold stratified cross validation*
 - c. Fit the best model
2. Built-in GridSearchCV Implementation: *run_built_in_grid_search*

Introduction

This report details the work performed in the Week 4 Machine Learning lab, which focused on building a complete machine learning pipeline for model selection and evaluation. The primary tasks involved implementing hyperparameter tuning and ensemble methods using two different approaches: a manual implementation and a more efficient, built-in scikit-learn implementation. This lab was designed to provide hands-on experience with these critical machine learning techniques and to compare the results of the two implementation methods.

Dataset Description

For this lab, the complete pipeline was run on the HR Attrition dataset. This dataset contains a variety of work-related and personal factors to predict whether an employee will leave the company (attrition). The notebook included functions to load and preprocess the dataset. The HR Attrition dataset was successfully loaded and preprocessed, resulting in:

	Instances	Features
Training set:	1029	46
Testing Set:	441	46

Methodology

1. *Hyperparameter Tuning:*

The process of finding the optimal set of hyperparameters for a machine learning model to achieve the best performance, where the hyperparameters are those that directly affect the performance of the model but aren't learnt by itself.

2. *Grid Search:*

A tuning method that systematically works through multiple combinations of hyperparameter values, evaluates a model for each combination, and identifies the best-performing one.

3. *k-Fold Cross-Validation:*

A robust technique for estimating a model's performance on unseen data by splitting the dataset into k folds. The model is trained on $k-1$ folds and validated on the remaining fold, with this process repeated k times. This method was used with 5 folds in this lab.

4. ML Pipeline:

- a. *StandardScaler*:
Standardizes features to have a mean of 0 and a standard deviation of 1.
- b. *SelectKBest*:
Selects the top k features based on a statistical test, where k is a hyperparameter to be tuned.
- c. *Classifier*:
The final model, which was one of three fundamental classification algorithms: Decision Tree, k-Nearest Neighbors (kNN), or Logistic Regression.

5. Implementation Process:

- a. Manual Grid Search Implementation: *run_manual_grid_search*
 - a. Define Parameter Grids
 - b. Implement Search Loop – Calculate avg ROC, AUC score for *5-fold stratified cross validation*
 - c. Fit the best model
- b. Built-in GridSearchCV Implementation: *run_built_in_grid_search*

Model	Best Parameters	Avg CV Score (ROC AUC)
Decision Tree (Manual)	{'classifier__criterion': 'entropy', 'classifier__max_depth': 5, 'classifier__min_samples_split': 10, 'feature_selection_k': 10}	0.7226
k-Nearest Neighbors (Manual)	{'classifier__n_neighbors': 9, 'classifier__weights': 'distance', 'feature_selection_k': 10}	0.7226
Logistic Regression (Manual)	{'classifier__C': 0.1, 'classifier__penalty': 'l2', 'feature_selection_k': 'all'}	0.8328

Model	Accuracy	Precision	F1-Score
<i>Manual Implementation</i>			
Decision Tree	0.8345	0.4706	0.3048
k-Nearest Neighbors	0.8186	0.3784	0.2593
Logistic Regression	0.8798	0.7368	0.5138
<i>Built-in GridSearchCV</i>			
Decision Tree	0.8345	0.4706	0.3048
k-Nearest Neighbors	0.8186	0.3784	0.2593
Logistic Regression	0.8798	0.7368	0.5138

Output Screenshots

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#####
PROCESSING DATASET: HR ATTRITION
#####
IBM HR Attrition dataset loaded and preprocessed successfully.
Training set shape: (1029, 46)
Testing set shape: (441, 46)
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RUNNING MANUAL GRID SEARCH FOR HR ATTRITION
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--- Manual Grid Search for Decision Tree ---
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Best parameters for Decision Tree: {'classifier__max_depth': 5, 'classifier__min_samples_split': 10, 'classifier__criterion': 'entropy', 'feature_selection_k': 10}
Best cross-validation AUC: 0.7226
--- Manual Grid Search for k-Nearest Neighbors ---
-----
Best parameters for k-Nearest Neighbors: {'classifier__n_neighbors': 9, 'classifier__weights': 'distance', 'feature_selection_k': 10}
Best cross-validation AUC: 0.7226
--- Manual Grid Search for Logistic Regression ---
-----
Best parameters for Logistic Regression: {'classifier__C': 0.1, 'classifier__penalty': 'l2', 'feature_selection_k': 'all'}
Best cross-validation AUC: 0.8328
=====
EVALUATING MANUAL MODELS FOR HR ATTRITION
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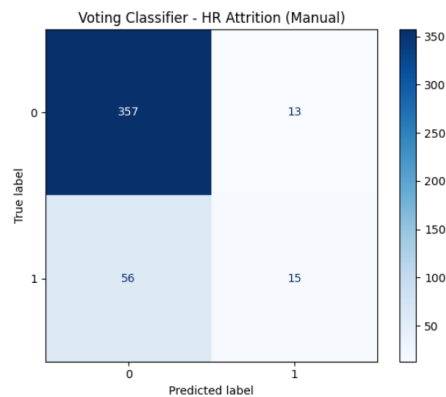
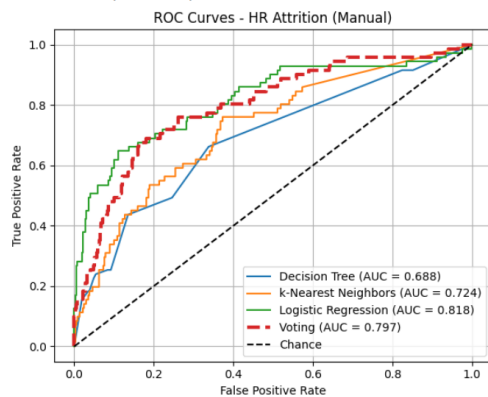
--- Individual Model Performance ---

Decision Tree:
Accuracy: 0.8345
Precision: 0.4706
Recall: 0.2254
F1-Score: 0.3048
ROC AUC: 0.6879

k-Nearest Neighbors:
Accuracy: 0.8186
Precision: 0.3784
Recall: 0.1972
F1-Score: 0.2593
ROC AUC: 0.7236

Logistic Regression:
Accuracy: 0.8798
Precision: 0.7368
Recall: 0.3944
F1-Score: 0.5138
ROC AUC: 0.8177

--- Manual Voting Classifier ---
Voting Classifier Performance:
Accuracy: 0.8435, Precision: 0.5357
Recall: 0.2113, F1: 0.3030, AUC: 0.7971
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RUNNING BUILT-IN GRID SEARCH FOR HR ATTRITION
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--- GridSearchCV for Decision Tree ---
Fitting 5 folds for each of 72 candidates, totalling 360 fits
Best params for Decision Tree: {'classifier__criterion': 'entropy', 'classifier__max_depth': 5, 'classifier__min_samples_split': 10, 'feature_selection_k': 10}
Best CV score: 0.7226

--- GridSearchCV for k-Nearest Neighbors ---
Fitting 5 folds for each of 24 candidates, totalling 120 fits
Best params for k-Nearest Neighbors: {'classifier__n_neighbors': 9, 'classifier__weights': 'distance', 'feature_selection_k': 10}
Best CV score: 0.7226

--- GridSearchCV for Logistic Regression ---
Fitting 5 folds for each of 9 candidates, totalling 45 fits
```

```
Best params for Logistic Regression: {'classifier__C': 0.1, 'classifier__penalty': 'l2', 'feature_selection__k': 'all'}
Best CV score: 0.8328
```

EVALUATING BUILT-IN MODELS FOR HR ATTRITION

--- Individual Model Performance ---

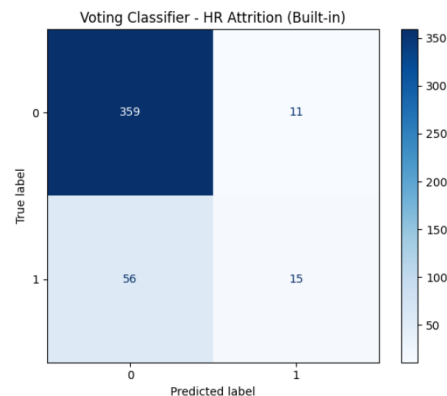
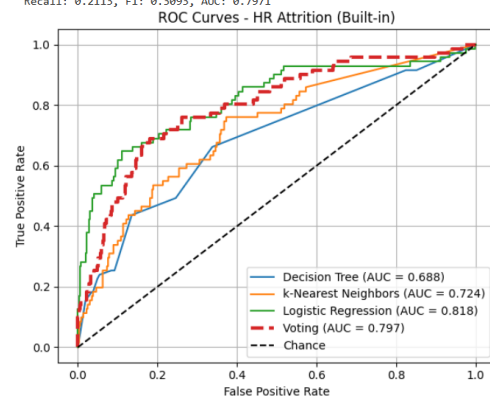
Decision Tree:
Accuracy: 0.8345
Precision: 0.4706
Recall: 0.2254
F1-Score: 0.3048
ROC AUC: 0.6879

k-Nearest Neighbors:
Accuracy: 0.8186
Precision: 0.3784
Recall: 0.1972
F1-Score: 0.2593
ROC AUC: 0.7236

Logistic Regression:
Accuracy: 0.8798
Precision: 0.7368
Recall: 0.3944
F1-Score: 0.5138
ROC AUC: 0.8177

--- Built-in Voting Classifier ---

Voting Classifier Performance:
Accuracy: 0.8481, Precision: 0.5769
Recall: 0.2113, F1: 0.3093, AUC: 0.7971



Completed processing for HR Attrition

ALL DATASETS PROCESSED!

Conclusion

The manual implementation of the grid search helped provide a good grasp of the inner workings of cross-validation and hyperparameter tuning. We were able to verify our custom implementation and recognize the effectiveness and convenience of utilizing a strong library by contrasting the manual results with those from scikit-learn's optimized GridSearchCV. The HR Attrition dataset analysis showed that the best classifier for this particular issue was Logistic Regression, underscoring the significance of assessing and contrasting various models in order to reach reliable conclusions. All things considered, this lab reaffirmed how important model evaluation and selection are to the machine learning pipeline.